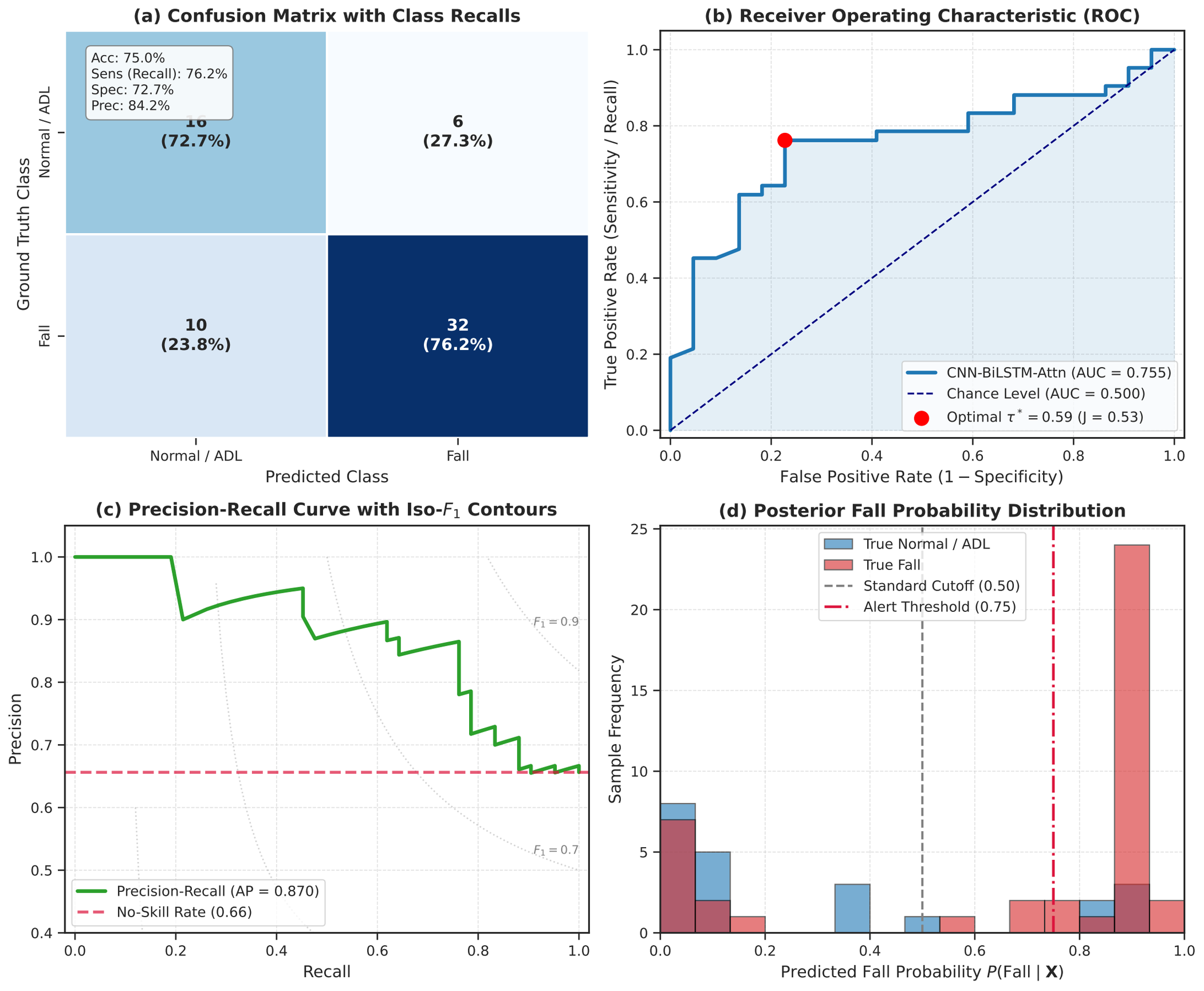
(a) Confusion Matrix with Class Recalls
Acc: 75.0%
Sens (Recall): 76.2%
Spec: 72.7%
Prec: 84.2%
16
(72.7%)
6
(27.3%)
10
(23.8%)
32
(76.2%)
Normal / ADL
Fall
Ground Truth Class
Normal / ADL
Fall
Predicted Class
(b) Receiver Operating Characteristic (ROC)
True Positive Rate (Sensitivity / Recall)
False Positive Rate (1 − Specificity)
CNN-BiLSTM-Attn (AUC = 0.755)
Chance Level (AUC = 0.500)
Optimal τ* = 0.59 (J = 0.53)
(c) Precision-Recall Curve with Iso-F1 Contours
Precision
Recall
F1 = 0.9
F1 = 0.7
Precision-Recall (AP = 0.870)
No-Skill Rate (0.66)
(d) Posterior Fall Probability Distribution
Sample Frequency
Predicted Fall Probability P(Fall | X)
True Normal / ADL
True Fall
Standard Cutoff (0.50)
Alert Threshold (0.75)